Family in Macroeconomics
Introduction

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GRIPS

Spring 2024

Family in Macro: What Fits In?

- Research at the intersection of macroeconomics and family economics
- Covers essentially anything related to family and household decisions and their impact on macroeconomy
- Why family and macro?
  - Many of the key decisions in macroeconomic models (labor supply, consumption, saving, human capital investment, fertility, etc) occur within a family.
  - Yet a typical macro model ignores family and within-household heterogeneity.

Family in Macro: What Fits In?

- Household decisions are central in macroeconomy and critically behind movements of macro variables
  - family formation: fertility, marriage and divorce
  - labor supply of men and women, roles of family insurance, effects on business cycles
  - time use and home production, household technology
  - children's education
  - cares for the elderly
  - intergenerational and inter-vivos linkage, transfers and bequests
  - inequality across genders

Family in Macro: What Fits In?

- Clearly important factors at an individual level, but also at the macro level. Household decisions formulate the trend and fluctuations of aggregate labor supply, productivity and growth, inequality, national saving, and so on.
- Family decisions ought to be an integral part of policy analysis.
- But: family macro is rather a new and under-explored field despite the importance. Why so?
  - Computational challenge, data limitation
  - Both are less of an issue now.

Family in Macro

- Excellent overview/surveys and a book
  - Doepke and Tertilt (2016) "Families in Macroeconomics," Handbook of Macroeconomics
  - Greenwood, Guner and Marto (2023) "The Great Transition: Kuznets Facts for Family-Economists," Handbook of the Economics of the Family
  - Doepke, Hannusch, Kindermann and Tertilt (2023) "The Economics of Fertility: A New Era," Handbook of the Economics of the Family
  - Greenwood (2018) *Evolving Households: The Imprint of Technology on Life*, MIT Press
- First warm-up assignment
  - Interactive family facts (US and cross-country) https://www.ricardomarto.com/data/
  - Identify an interesting fact about family. Ready to give a 5-minute presentation to tell us about the fact, why you find it interesting, (if possible, background, potential theory and more data behind the fact)

Introduction: Becker and Early Models

- See also Doepke (2015, J of Demog Econ)
- Pre-Becker
  - Family, fertility $\not\Leftrightarrow$ economics
  - "Taste" for children varying over time and by income levels
- Becker (1960)
  - Consider a child as an economic commodity, chosen based on income and relative price, under a *fixed* preference.
  - A *concept* of a quantity-quality trade-off. Quality to account for strong income effects (fertility rises at the high end of income distribution): knowledge of birth control correlated with income, driving negative effects.

Introduction: Becker and Early Models

- Becker and Lewis (1973)
  - Formalization of the quantity-quality trade-off, connected via the budget constraint. Positive income effect vs negative substitution effect to account for income-fertility relationship.
- Becker and Tomes (1976)
  - Extend the Becker-Lewis model with child "endowment" consisting of inherited ability, public education, etc.
- So far theories of fertility choice, nothing to do with macro or growth.

Introduction: Becker and Early Models

- Becker and Barro (1988)
  - Extend the Becker-Lewis model with an explicit intergenerational linkage via parental altruism. Child quality represents the child's utility, rather than the abstract quality enjoyed by parents.
  - Interest rate positively correlated with fertility via substitution effects (partial eq). Barro and Becker (1989) add general eq.
- Becker, Murphy and Tamura (1990)
  - Extend the Barro-Becker model with human capital as the engine of growth, as in Lucas (1988).
  - Productivity consists of innate ability and acquired human capital. Possibility of multiple equilibria, one with high growth and low fertility, another with low growth and high fertility.

Introduction

- Doepke and Tertilt (2016, HB of macroeconomics 2B, Ch 23)
- Early literature focused on the roles of home production in business cycle models to account for cyclicality of investment and labor supply (Greenwood and Hercowitz 1991, Benhabib, Rogerson and Write 1991).
- More recent literature (models covered in this chapter) is focused on interactions within family: horizontal (husband and wife) and vertical (parents and children)

Outline

- The HB chapter reviews some family facts and presents various models of family and macro to account for the family decisions in the short, medium and long-run.
- **Section 2:** Family and Macro in the Short and Medium Run
  - Recent changes in family structure in the U.S. Models to account for the trend facts.
  - Family's labor supply. Micro and macro elasticity. Insurance roles of family's labor supply against idiosyncratic risks (micro) and aggregate shocks (macro).
  - Family's saving, effects of divorce.

Outline

- **Section 3:** Family and Economic Growth
  - Family facts in the long-run
  - Models of intergenerational linkage, fertility, men and women. Roles of human capital for growth, interacting with family decisions.
- **Section 4:** Family and Political Economy, Development and Institutional Change
  - Long-run economic development and family. Changes in institutions such as democracy (voting rights of women), child labor, compulsory education system, linked to family structures and labor market inequality.

Representative Agent Models and the Departure

- A prototype RBC model

$$\max_{\{c_t, l_t\}_{t=0}^{\infty}} E\left\{\sum_{t=0}^{\infty} \beta^t U(c_t, l_t)\right\}$$

subject to

$$\begin{aligned} c_t + a_{t+1} &= w_t l_t + (1 + r_t) a_t \\ a_{t+1} &> -B \\ a_0 &= 0 \\ 0 &\leq l_t \leq T \end{aligned}$$

Representative Agent Models and the Departure

- RA models underlie many macro models in the RBC and DSGE literature and serve as simplified representation of the economy.
- There are, however, severe limitations if forces a RA model abstracts from substantially change overtime *and* affect macroeconomic behavior.
- Also, the prototype model has nothing to say about heterogeneity and inequality across households.
- Extensions of RA models to incorporate:
  - heterogeneity in age: OLG models
  - heterogeneity in income and wealth: models of inequality
  - heterogeneity across household members: **models of family**

Family Facts: Decline in Fertility

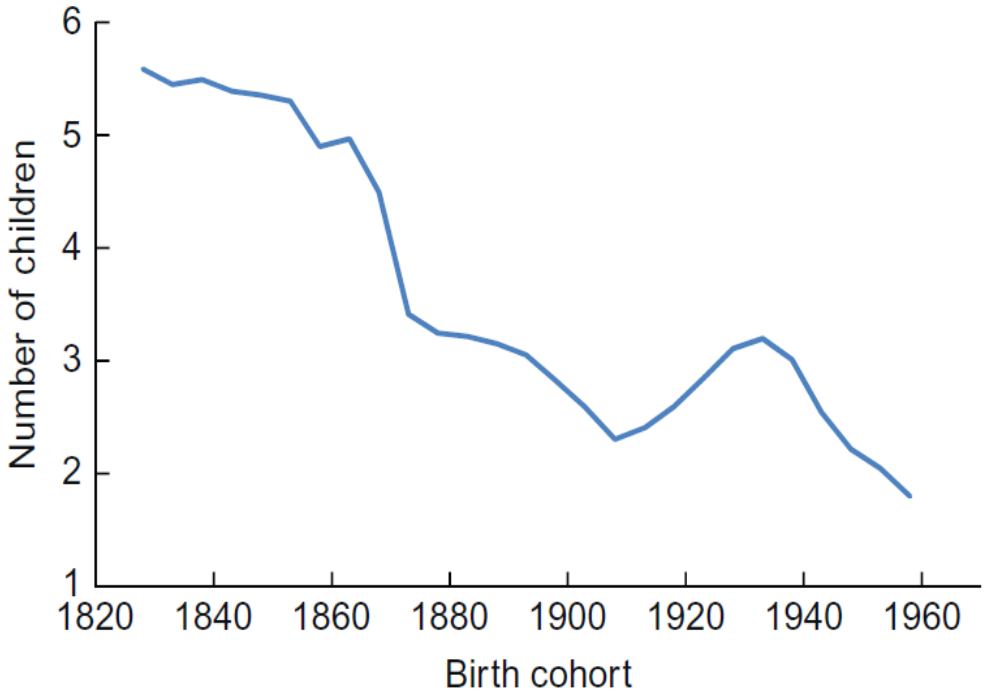


Fig. 1 Children ever born by cohort, United States (ie, average number of children for women born in a given year). *Jones, L.E., Tertilt, M., 2008. An economic history of the relationship between occupation and fertility—U.S. 1826–1960. In: Rupert, P. (Ed.), Frontiers of Family Economics, vol. 1. Emerald Group Publishing Limited, Bingley, UK (Table 1A).*

Family Facts: Decline in Household Size

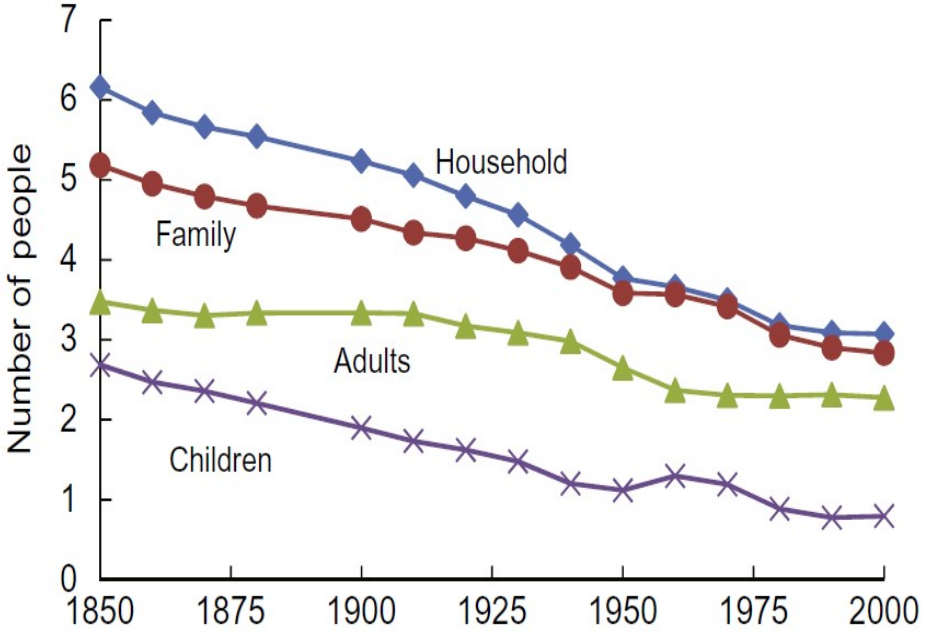


Fig. 2 Household size over time, United States. *Salcedo, A., Schoellmann, T., Tertilt, M., 2012. Families as roommates: changes in US household size from 1850 to 2000. Quant. Econ. 3 (1), 133–175 (Figure 1).*

Family Facts: Changes in the Marriage

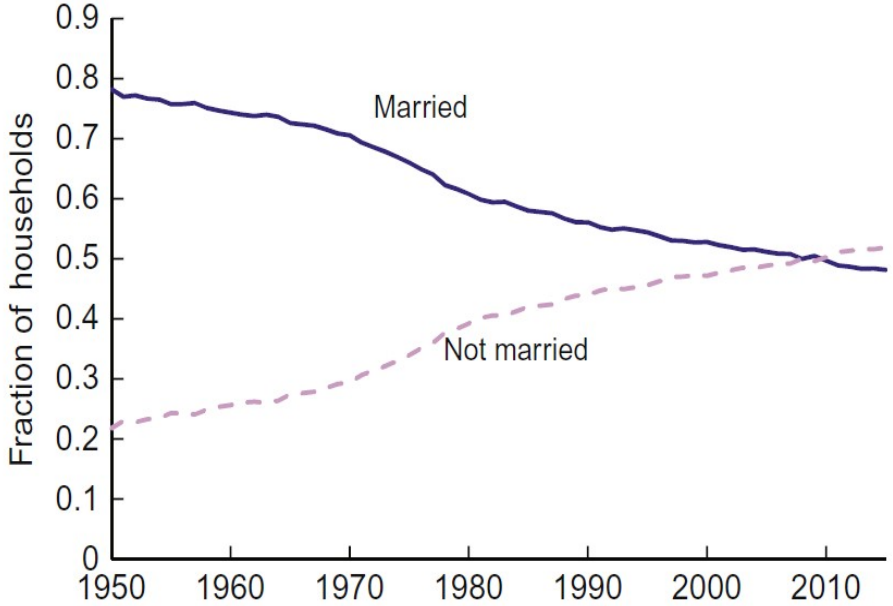


Fig. 3 Proportion of households including a married couple vs all other households over time, United States. *US Census Bureau, Historical Time Series, Current Population Survey, March and Annual Social and Economic Supplements, 2015 and earlier.*

Family Facts: Changes in the Single Households

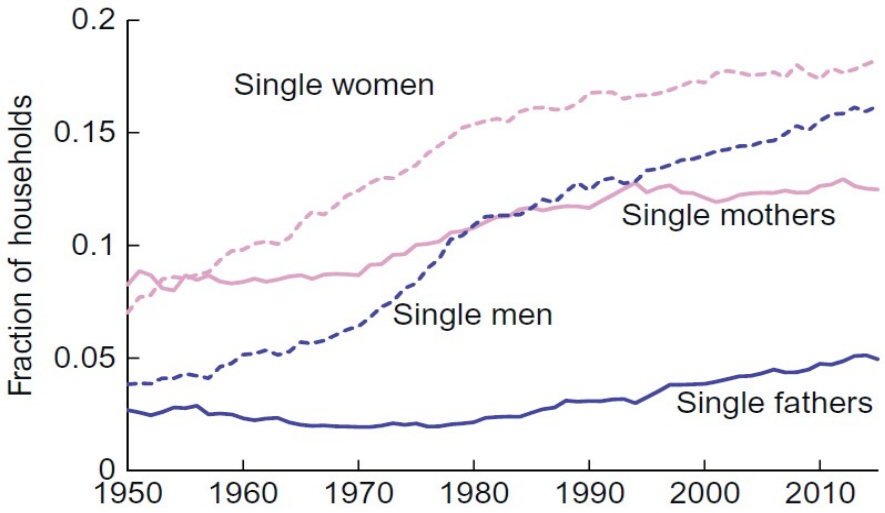


Fig. 4 Nonmarried households by type over time, United States. *US Census Bureau, Historical Time Series, Current Population Survey, March and Annual Social and Economic Supplements, 2015 and earlier.*

Family Facts: A Rise in Divorce

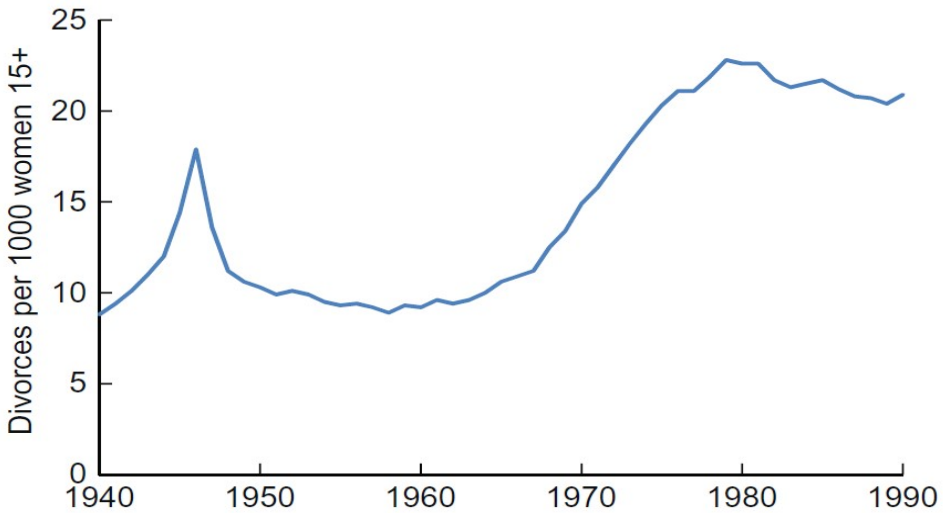


Fig. 6 Divorce rate over time, United States. *US Vital Statistics; Clarke, S.C., 1995. Advance report of final marriage statistics, 1989 and 1990. Monthly Vital Stat. Rep. 43 (12), 3–5.*

Family Facts: Changes in LFP

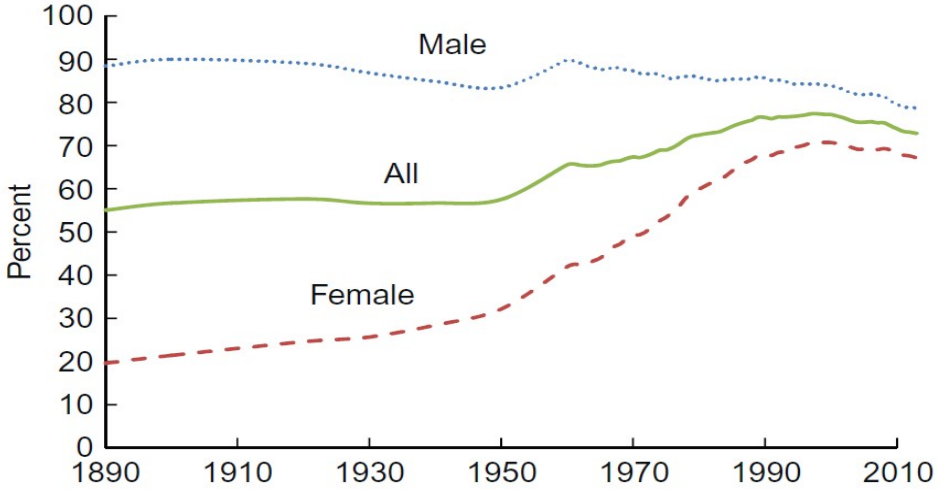


Fig. 7 Labor force participation by gender over time, all persons 16–64 years old, United States. *OECD LFS Sex and Age Indicators and US Department of Commerce, Bureau of the Census, "Historical Statistics of the United States: Colonial Times to 1970," Bicentennial Edition, Part 1, 1975, Tables A119–134 and D29–41.*

Family Facts: Changes in Gender Wage Gap

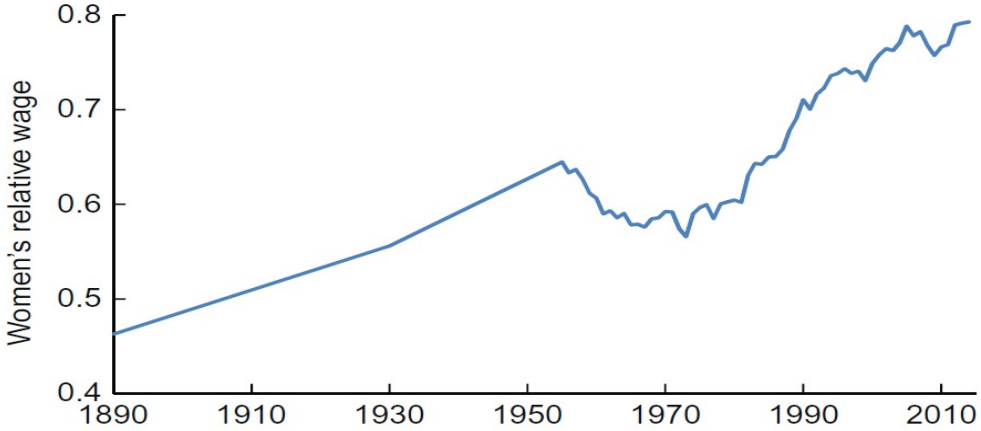


Fig. 10 Gender wage gap: median earnings of full-time, year-round, female workers 15 years and older, relative to men, United States. *US Census Bureau, Historical Income Tables. Numbers for 1890 and 1930 are from Goldin, C., 1990. Understanding the Gender Gap: An Economic History of American Women. Oxford University Press, Oxford (Table 3.2).*

Explaining Facts

- Growing literature to explain facts about family transformations using structural models, for example:
- Secular decline in fertility
  - Variations of the quality-quantity trade-off model (Becker and Lewis, 1973) with a rising return to education due to technological progress
- Secular increase in female labor force participation
  - Development of home production technology
  - Technological advances in health (infant formula, etc)
  - A decline in the gender wage gap (education, technological and industrial changes, etc)
- Interaction of the two (fertility and female LFP)
  - Correlation + or −?
  - Roles of childcare facilities and policies, roles of men,...
- Causes of marriage, a rise in divorce, single motherhood, cohabitation,...

Family and Aggregate Economy

- Review models to connect changing family structures and aggregate economy

  1. Family and aggregate labor supply

     - joint labor supply decision of household members

     - endogenous bargaining

  2. Family and aggregate savings

  3. Private information and household decisions

Models of Families

- How to model a decision of a family consisting of multiple adult members e.g. husband and wife?
- **Unitary models**
  - Distinguish between behavior of household members (e.g. labor supply of spouses), but assume a unified household-utility, rather than separate preferences for each
  - No bargaining between spouses
- **Non-unitary models**
  - Explicitly models conflict of interest between spouses (backed by separate utility functions) resolved through a bargaining process, cooperative or non-cooperative (private information).
  - Many papers use cooperative bargaining such as Nash bargaining subject to an outside option, but some empirical evidence against cooperative model (see papers by Udry, for example).

Family and Aggregate Labor Supply

- Consider joint labor supply decision of men and women in the extensive margin.
- People can be either single or married (or cohabiting).
- The utility function of an individual of gender g is given as

$$U_g(c_g, l_g) = \log(c_g) - \eta_g l_g$$

  - c_g consumption
  - $l_g \in \{0, 1\}$ labor supply
  - η_g relative weight on leisure

Budget Constraint

- Singles

$$c_g + \psi l_g = w_g l_g + \gamma_g$$

  - $0 < \psi < \min(w_f, w_m)$: fixed cost of running a household conditional on working
  - γ_g: unearned income (endowment or transfer income)

- Married couples

$$c_f + c_m + \psi \min(l_f, l_m) = w_f l_f + w_m l_m + \gamma \tag{1}$$

where $\gamma = \gamma_f + \gamma_m$, sum of a couple's unearned income.

Labor Supply Decisions

- Singles work if

$$\log(w_g + \gamma_g - \psi) - \eta_g \geq \log(\gamma_g)$$

or

$$\eta_g \leq \log\left(\frac{w_g + \gamma_g - \psi}{\gamma_g}\right) \tag{2}$$

  - More likely to work if work disutility (LHS) is not so large relative to the opportunity cost of leisure (RHS): w_g is high, γ_g is *low*, ψ is low, or η_g is low

Labor Supply Decision

- For a married couple, assume a household solves a problem with exogenous bargaining weights λ_g s.t. $\lambda_f + \lambda_m = 1$
- A married couple's problem

$$\max\{\lambda_f[\log(c_f) - \eta_f l_f] + \lambda_m[\log(c_m) - \eta_m l_m]\} \tag{3}$$

  subject to (1).
- In equilibrium, $c_g = \lambda_g c$, where c is the total consumption of a household.
- Denote by $V(l_f, l_m)$ the value of (3) for given labor supply and conditional consumption allocation.
- Assume for simplicity that a husband always works as long as $w_m > 0$.
- The wife will work if $V(l_f = 1, 1) \geq V(l_f = 0, 1)$.

Labor Supply Decision

- Women work if and only if

$$\eta_f \leq \frac{1}{\lambda_f} \log\left(\frac{w_f + w_m + \gamma - \psi}{w_m + \gamma}\right) \equiv \frac{1}{\lambda_f} \log\left(\frac{c_1}{c_0}\right)$$

  where c_1 is the family consumption if $l_f = 1$ and c_0 if $l_f = 0$.

- Women more likely to work with low bargaining power λ_f. Also, if home production cost ψ is low, w_m is low, and w_f is high.

- Could the value of marriage fall below the value of singles (e.g. low λ_f case)? $\rightarrow$ endogenous bargaining power to satisfy the participation constraints (later)

Labor Supply Elasticity

- Now study labor supply elasticity (of the extensive margin) for singles and married couples.
- Consider heterogeneity in $\eta_g \sim F(\eta_g)$, $f(\eta_g) = F'(\eta_g)$. Participate if η_g is below the threshold (reservation property).
- For singles, from (2), the fraction of participants N_g^s is given as

$$N_g^s = F\left(\log\left(\frac{w_g + \gamma_g - \psi}{\gamma_g}\right)\right)$$

- Similarly for married couples

$$N_g^m = F\left(\frac{1}{\lambda_f}\log\left(\frac{w_f + w_m + \gamma - \psi}{w_m + \gamma}\right)\right)$$

Labor Supply Elasticity

- Aggregate wage elasticity for singles

$$\xi^s \equiv \frac{\partial N_g^s}{\partial w_g}\frac{w_g}{N_g^s} = \frac{w_g}{w_g + \gamma_g - \psi}\frac{F'}{F}$$

- For married couples

$$\xi^m \equiv \frac{\partial N_f^m}{\partial w_f}\frac{w_f}{N_f^m} = \frac{w_f}{\lambda_f(w_f + w_m + \gamma - \psi)}\frac{F'}{F}$$

- Can show $\xi^m > \xi^s$, i.e. **married women's labor supply is more elastic**, if women's unearned income γ_f is sufficiently small, in line with micro estimates (e.g. Blundell and MaCurdy 1999)

Insurance Roles of Family

- So far, only own-wage elasticity of female labor supply. Now consider insurance roles of labor within a family
- Extend the model with unemployment shocks
  - With probability u an individual is unemployed (or wage falls to zero), independently of marital status and gender
  - Now some husbands may or may not be working due to unemployment shocks

Insurance Roles of Family

- Singles' aggregate labor supply

$$N_g^s = (1-u)F\left(\log\left(\frac{w_g + \gamma_g - \psi}{\gamma_g}\right)\right)$$

- Elasticity of labor supply with respect employment $(1-u)$, ν^s, is 1.
  - The threshold η_g is independent of employment risks

Insurance Roles of Family

- For married couples, the threshold values of η_f depends on the employment status of the husband.

$$\begin{aligned}\hat{\eta}_e &= \frac{1}{\lambda_f}\log\left(\frac{w_f + w_m + \gamma - \psi}{w_m + \gamma}\right)\\ \hat{\eta}_u &= \frac{1}{\lambda_f}\log\left(\frac{w_f + \gamma}{\gamma}\right)\end{aligned}$$

- $\hat{\eta}_e < \hat{\eta}_u$, that is, wives of unemployed husbands work with strictly higher probability than wives of employed husbands
  - ψ need not be paid if a husband is unemployed: **substitution effect**
  - household income is lower with an unemployed husband: **income effect**
  - both work in the same direction

Insurance Roles of Family

- Aggregate labor supply of married couples

$$N^m = (1-u)[1+(1-u)F(\hat{\eta}_e)] + u(1-u)F(\hat{\eta}_u)$$

- Elasticity of married labor supply with respect to $(1-u)$

$$\nu^m \equiv \frac{\partial N^m}{\partial(1-u)}\frac{1-u}{N^m} = \frac{1+2(1-u)F(\hat{\eta}_e)-(1-2u)F(\hat{\eta}_u)}{1+(1-u)F(\hat{\eta}_e)+uF(\hat{\eta}_u)}$$

  - Since $\hat{\eta}_u > \hat{\eta}_e$ and $F(\hat{\eta}_u) > F(\hat{\eta}_e)$, can show the elasticity $\nu^m < 1 = \nu^s$
  - The fraction of women, $F(\hat{\eta}_u) - F(\hat{\eta}_e)$, do not work if the husband is employed but choose to work if he is unemployed.

- Insurance in the family dampens the fluctuations in aggregate employment. Married women's labor supply is more elastic at the microlevel, but their labor supply dampens the volatility of aggregate labor supply via the family insurance effect.

Endogenous Bargaining

- So far assumed full commitment and exogenous bargaining
  - If female bargaining power λ_f is low and w_f is high, such women are likely to work a lot and consume little. But they'd rather leave the marriage were it not for the full commitment!
  - Efficient bargaining would dictate bargaining weights to adjust so women are at least as well off as they would be if they were single.
  - Such adjustments are possible as long as the surplus of marriage is positive ($\psi > 0$ in the current model).
    - Positive surplus could arise by other channels such as consumption scale and equivalence, positive utility of marriage, utility of children, etc.

Endogenous Bargaining (A Sketch)

- Consider a case where the bargaining weight adjusts to a change in w_g.
- Suppose a woman always prefers to work (for simplicity of exposition).
- A woman always prefers to be single if

$$\log(w_f + \gamma_f - \psi) - \eta_g > \log(\lambda_f(w_f + w_m + \gamma_f + \gamma_m - \psi)) - \eta_g$$

or

$$w_f + \gamma_f > \frac{\lambda_f}{1 - \lambda_f}(w_m + \gamma_m) + \psi$$

- The condition holds if w_f is high or λ_f is low. She'd stay married if the bargaining weight adjust to guarantee her at least the reservation utility:

$$\lambda_f = \frac{w_f + \gamma_f - \psi}{w_f + w_m + \gamma_f + \gamma_m - \psi}$$

Endogenous Bargaining (A Sketch)

- In a dynamic model, can consider the adjustment of λ_f in response to a shock to the wages w_g, unearned income γ_g, and participation cost η_f to keep a wife in marriage. Also ψ, which affects participation cost of both men and women.

- What about the effects of endogenous bargaining on aggregate labor supply elasticity?

  - If a rise in relative wage increases her bargaining power, it also increases the weight on her leisure, which lowers labor supply response. But it also affects the other spouse in an opposite way, hence net effect is ambiguous.

  - See Knowles (2013) and Voena (2015), and also the chapter for discussions on empirical evidence.

Family Labor Supply and Jobless Recoveries

- Post-1990, recoveries from a recession have been accompanied by a sluggish increase in employment
  - Structural change and a rise in job polarization, fixed cost of labor adjustment, etc.
  - Roles of women's participation trend (Albanesi 2014): a model with gender-biased technological change to account for the long-run trend of a rise in FLFP and jobless recovery.

Table 2 Jobless recoveries: change in employment/population ratio in 4 years after peak in unemployment rate, in percentage points, by gender (includes three pre-1990 and tree post-1990 recessions)

	Change in E/P		
Period	**Total**	**Men**	**Women**
Pre-1990	0.65	−2.62	5.85
Post-1990	−2.78	−3.94	−1.41
Post-1990, excl. Great Recession	−1.35	−2.47	−0.07

Notes: Pre-1990 recessions include the 1969, 1973, and 1981 recessions. Post-1990 recessions include the 1990, 2001, and 2007 recessions.

Families and Saving

- Channels through which family affects saving
  - Changes in the household size over the life-cycle
  - Precautionary savings against fluctuations of family income and expenditures
  - New sources of risk associated with family structure via marriage, divorce and fertility

Savings and Divorce: A Simple Model (A Sketch)

- Two-period model. Individuals are married in period 1. A divorce occurs with probability π in period 2.
- Divorce regime (policy): a wife retains a κ_f fraction of household assets and a husband $\kappa_m = (1 - \kappa_f)$.
- Assume both husband and wife work. They choose family saving a' and consumption of each, c_f and c_m, under cooperative bargaining with fixed bargaining weight of λ_f and λ_m.

Savings and Divorce

- The couple's decision in period 1

$$\max_{c_f, c_m, a'} \Big\{ \lambda_f \log(c_f) + \lambda_m \log(c_m)$$
$$+\beta[\lambda_f(\pi V_f^D(a') + (1-\pi)V_f(a')) + \lambda_m(\pi V_m^D(a') + (1-\pi)V_m(a'))] \Big\}$$

subject to

$$c_f + c_m + a' = w_f + w_m$$

where V_g^D is the value in the case of divorce given as

$$V_g^D(a') = \log(w_g' + (1+r)\kappa_g a')$$

and V_g represents the value if marriage continues, given as

$$V_g(a') = \log[\lambda_g(w_f' + w_m' + (1+r)a')]$$

Savings and Divorce

- Effects of divorce on saving depends on the divorce regime (the property division rule) and the relative bargaining power.
- Can show that the presence of divorce risk increases savings, if for a spouse who is worse off by divorce, $\kappa_g > \lambda_g$. Intuition: additional saving would lower inequality, which generates additional demand for precautionary saving.
- See the chapter for a model with divorce and endogenous bargaining power.

Family and Economic Growth

1. Family + growth facts
2. Models of family and growth
   - A simple model of human capital (no family)
   - + parent and child and education (human capital inv.) via warm-glow
   - + fertility decisions
   - + two parents (men and women) with heterogeneity in
     - Preference (bargaining weight and altruism)
     - Technology (weight in production function, human capital transmission)
     - Endogenous bargaining (skill-dependent weight)

Family+Growth (1) Fertility and GDP

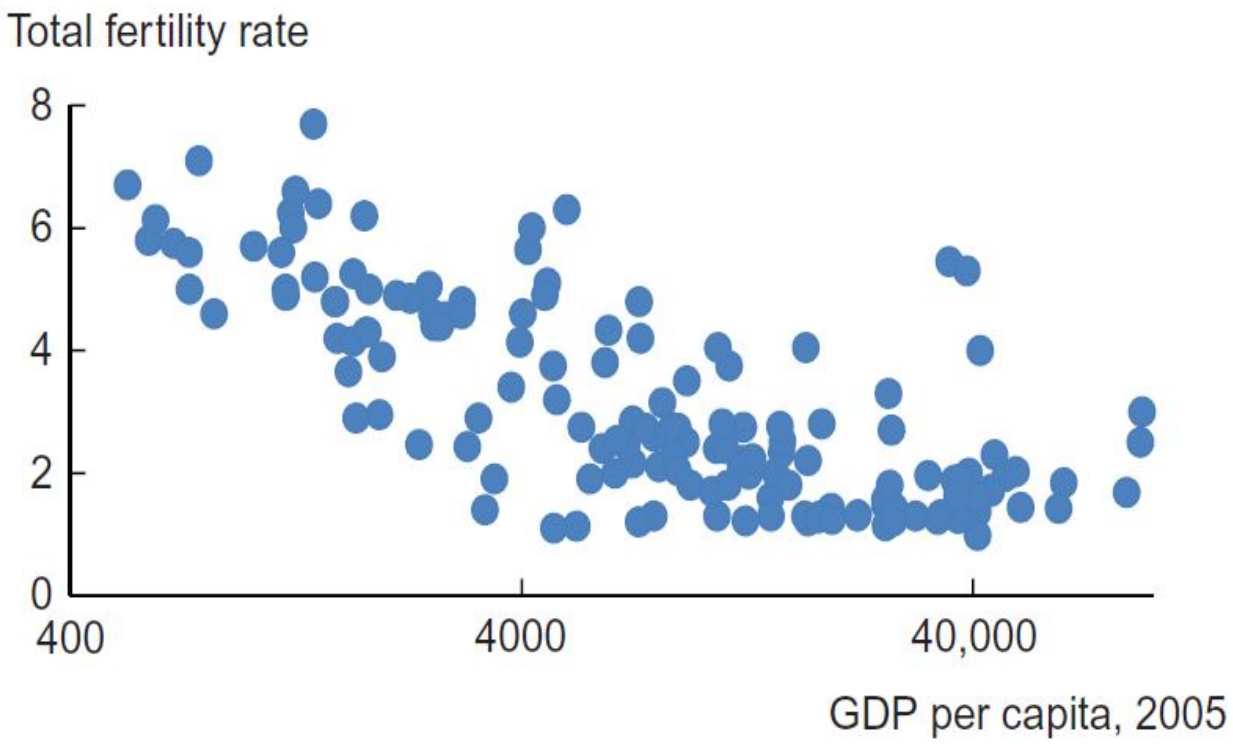


Fig. 17 TFR and GDP
Data: Cross-country data from GID (Gender, Institutions and Development Database, OECD) and World Development Indicators (WDI) 2005

Family+Growth (2) Fertility and Education

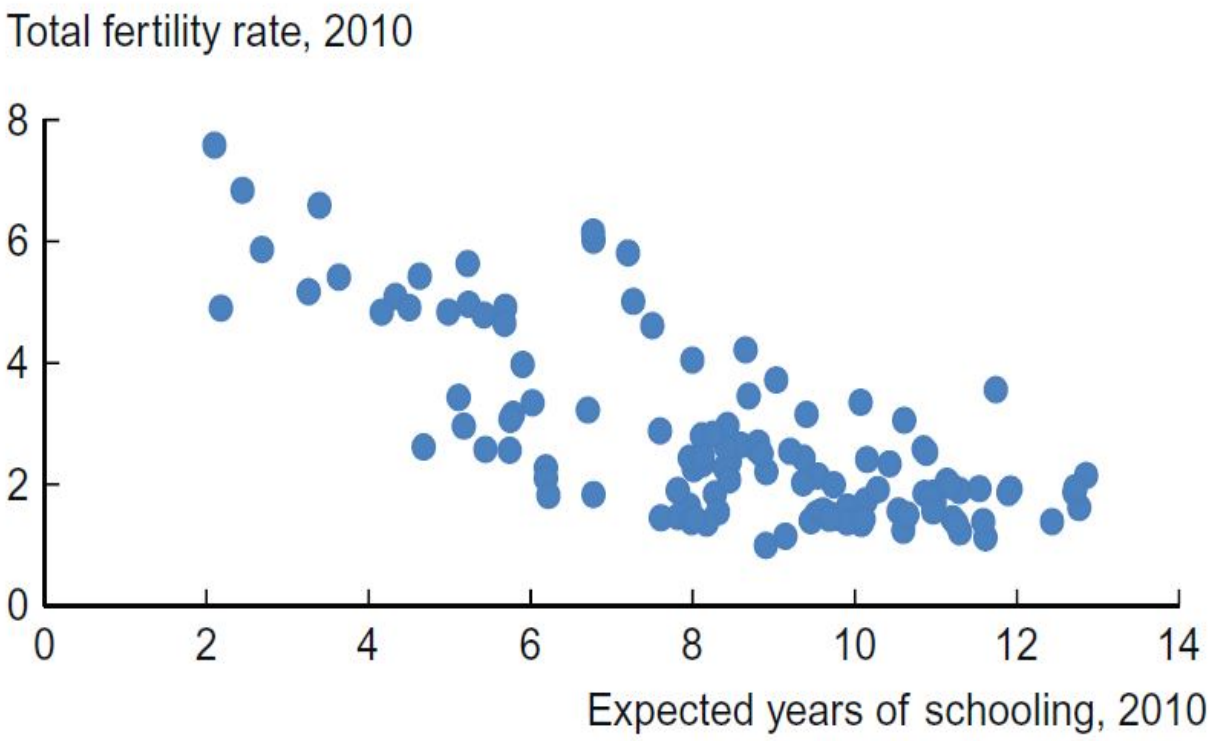


Fig. 18 TFR and Education
Data: World Development Indicators (WDI) data

Family+Growth (3) Early Marriage and GDP

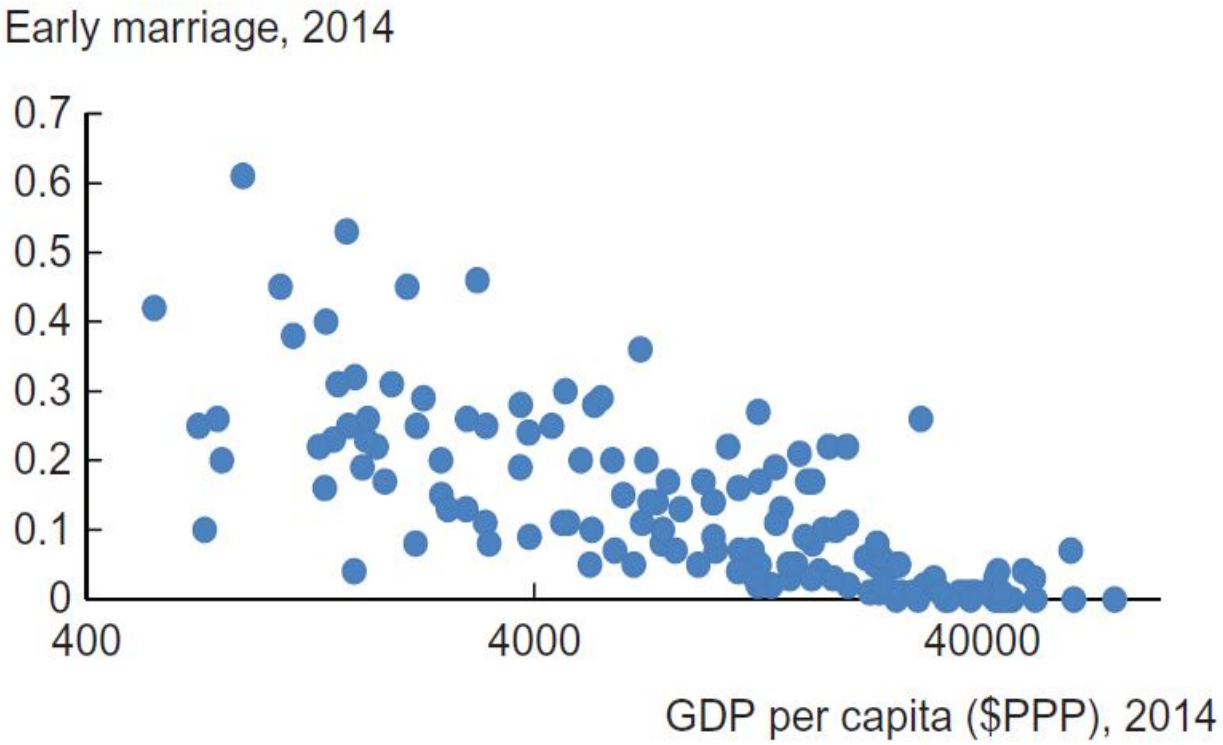


Fig. 19 Early Marriage (share of married at age 15-19)
Data: OECD Gender Statistics and WDI

Family+Growth (4) Female LFP and GDP

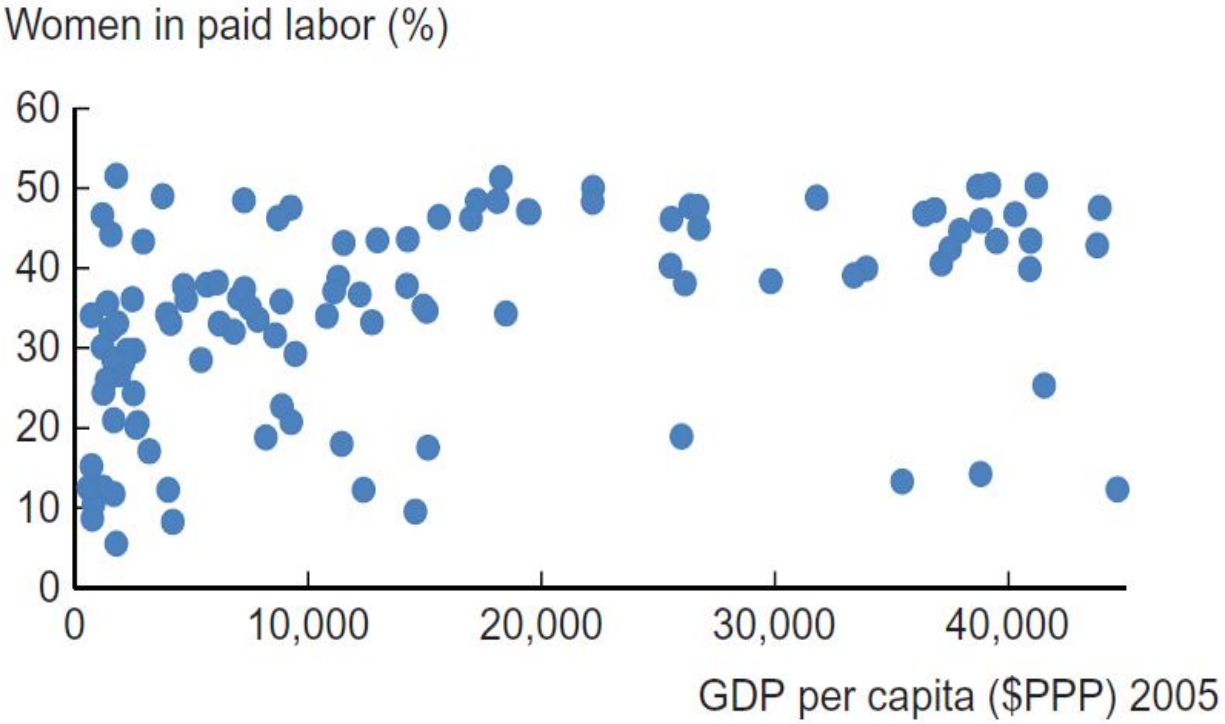


Fig. 20 Women in paid labor and GDP.
Data: OECD Gender Statistics and WDI

- Essentially all countries with GDPpc above $20,000, women make up more than 40% of paid labor force. (Rich countries with low participation rates are oil-rich countries.)

A Model of Human Capital

- Lucas (1988) model of human capital
- A unit time endowment is allocated to market work, u, and education (human capital investment), $1 - u$
- Production $Y = K^{\alpha}(uH)^{1-\alpha}$
  - K physical capital
  - H human capital
  - u work hours
- Human capital accumulation $H' = B(1 - u)H$
  - B technology, $(1 - u)$ time on education
  - Growth rate of human capital (gross), $B(1 - u) \longrightarrow$ next, consider $(1 - u)$ as a family decision s.t. growth is driven by endogenous accumulation of human capital and physical capital.

A Model of Parents/Children and Education

- Model of a parent and a child: add education decision of a parent via warm-glow bequest motives
- Exgenous fertility. No saving. Time resource allocation of a parent.
- Preferences

$$u(c) + \delta u(\gamma')$$

  - c a parent's consumption
  - δ a degree of warm-glow altruism
  - γ' a child's "full income"

A Model of Parents/Children and Education

- Consumption = production

$$c = AlH$$

  - A technology
  - l time for production
  - H human capital of a parent
  - "Full income" defined as income with $l = 1$

- Human capital of children

$$H' = (Be)^{\theta} H \tag{4}$$

  - B and θ technology parameters
  - e education time of a parent, $e + l \leq 1$

A Model of Parents/Children and Education

- A parent's problem is to choose e to

$$\max \{\log(c) + \delta\theta \log(e)\}$$

  where $c = AlH$ and $e + l = 1$.
- The optimal education

$$e^* = \frac{\delta\theta}{1 + \delta\theta}$$

  Plug it in (4), obtain growth rate of human capital (and consumption and output)

$$\frac{H'}{H} = \left(B\frac{\delta\theta}{1 + \delta\theta}\right)^{\theta}$$

- Remarks
  - Dependence of growth of human capital (and output and consumption) also on the degree of altruism δ, unlike in Lucas model
  - Degree of altruism δ need not be the same as the rate of time preference in a standard growth model
  - Intergenerational elasticity of substitution may differ from IES

A Model with Fertility Choice

- Now add fertility decisions. Assume one-parent family choosing the number of children
- The utility function

$$u(c) + \tilde{\delta} u(n) + \delta u(\gamma')$$

  - n number of children (choice)
  - $\tilde{\delta}$ preference parameter
  - γ' a child's full income (as before)
- Cost of raising children
  - ϕ fixed time cost per child (exogenous)
  - e education time per child (choice)
- Time constraint: $l + (\phi + e)n \leq 1$
- Assuming log-utility, the objective is to choose e and n to

$$\max \left\{ \log(c) + \tilde{\delta} \log(n) + \delta\theta \log(e) \right\}$$

A Model with Fertility Choice

- FOC for education e and fertility n

$$
\begin{aligned}
e^* &= \frac{\delta\theta}{\tilde{\delta} - \delta\theta}\phi \\
n^* &= \frac{\tilde{\delta} - \delta\theta}{\phi(1 + \tilde{\delta})}
\end{aligned}
$$

- The equilibrium growth rate of human capital

$$
\frac{H'}{H} = \left(B\frac{\delta\theta\phi}{\tilde{\delta} - \delta\theta}\right)^{\theta}
$$

- Remarks
  - Quantity vs quality tradeoff
    - Higher return to education θ raises education and reduces fertility
    - Higher fixed cost of children ϕ reduces fertility and raises education
  - High ϕ and high θ (that lower fertility) raise the growth rate H'/H. Same results as in Becker, et al (1990)
  - Two types of intergenerational preference parameters δ and $\tilde{\delta}$ affect the growth rate separately

A Model with Two Parents

- Finally, add two persons (parents) to the model and heterogeneity in
  - (1) Preference: bargaining weight and bequest motive
  - (2) Technology: weight in production function, human capital transmission
  - (3) Endogenous bargaining: skill-dependent weight

A Model with Two Parents (1) Preference

- Heterogeneity in preference: bargaining weight and bequest strength
- Exogenous fertility. No saving.
- Family's time cost of education.
- The objective function of a two-parent family is to choose education e to

$$\max \left\{\lambda[\log(c) + \delta_f \theta \log(e)] + (1-\lambda)[\log(c) + \delta_m \theta \log(e)]\right\}$$

  - λ represents the (fixed) bargaining weight of a woman, $(1-\lambda)$ the weight of a man
  - δ_f and δ_m represent the altruism of a woman and a man

A Model with Two Parents (1) Preference

- The optimal education

$$e^* = \frac{\tilde{\delta}\theta}{1 + \tilde{\delta}\theta}$$

where $\tilde{\delta} \equiv \lambda\delta_f + (1 - \lambda)\delta_m$

- The growth rate is

$$\frac{H'}{H} = \left(B\frac{\tilde{\delta}\theta}{1 + \tilde{\delta}\theta}\right)^{\theta}$$

  - They are the same as in the first model of a parent and children, replacing δ with $\tilde{\delta}$

- Remarks
  - Gender preference gaps matter for growth and family decisions
  - If $\delta_f > \delta_m$, the growth is higher if λ is larger
  - Note however female empowerment may REDUCE growth through capital investment channels (Doepke and Tertilt, 2014).
    If women like to spend more on human capital accumulation and men more on saving and physical capital, a money transfer to woman and a reallocation of capital may lower the growth, depending on the production function (and stages of development).

A Model with Two Parents (2) Technology

- Heterogeneity in technology: men and women differ in relative importance in (1) production and (2) transmission of human capital to children.
- Assume only women raise children and men work full time.
- Female time constraint $l_f + e_f + e_m \leq 1$
  - l_f market work
  - e_f and e_m time to educate daughters and sons
- Production function

$$c = A(l_f H_f)^{\alpha}(H_m)^{1-\alpha}$$

  α is the relative importance of women in production
- Children's human capital

$$\begin{aligned} H'_f &= (Be_f)^{\theta} H_f^{\beta} H_m^{1-\beta} \\ H'_m &= (Be_m)^{\theta} H_f^{\beta} H_m^{1-\beta} \end{aligned}$$

  β is the relative importance of a mother in transmitting human capital to children

A Model with Two Parents (2) Technology

- The objective is to choose labor supply l_f, and education, e_m and e_f, to

$$\max \left\{ \log(c) + \delta \left[\alpha\theta \log(e_f) + (1-\alpha)\theta \log(e_m) \right] \right\}$$

- From equilibrium allocations of l_f, e_m and e_f, can show

$$\frac{H_f}{H_m} = \left(\frac{e_f}{e_m} \right)^{\theta} = \left(\frac{\alpha}{1-\alpha} \right)^{\theta}$$

  - Equilibrium education is the ratio of relative importance in production.
  - Note no dependence on β, due to the warm-glow assumption. In a pure altruistic model, parents care about the effects of the education on grandchildren via children's human capital. Hence relative importance of men/women in child development does matter.

A Model with Two Parents (2) Technology

- Equilibrium growth rate of human capital (and output and consumption)

$$\frac{H'}{H} = \left\{ \frac{B\delta\theta}{\alpha + \delta\theta} (1-\alpha)^{1-\beta} \alpha^{\beta} \right\}^{\theta}$$

- The wage ratio per unit of time (gender wage gap) $w_f / w_m = \alpha / (1-\alpha)$
- Remarks
  - Higher δ (more altruism) $\rightarrow$ higher growth (same as before)
  - Differential roles of women α and β matter for growth, interacting with each other
  - Can show that the growth is hump-shaped in α
    - at low α, higher $\alpha \rightarrow$ a reduction in gender gap in education and wage, a rise in female LFP, and acceleration of growth
    - at high α, higher $\alpha \rightarrow$ cost of education too high, lower education investment and human capital of children
    - empirically, non-monotonic relationship between development and participation, cross country and over time.

A Model with Two Parents (3) Endogenous Bargaining

- Endogenize the bargaining weight λ in the education decision, as a function of gender education gap
  - note education is also endogenously chosen by the family
- Use a model that combines gender difference in

(1) preference (bargaining weight λ and altruism δ_f)

(2) technology (production α and human capital transmission β)

A Model with Two Parents (3) Endogenous Bargaining

- First, consider an **exogenous** bargaining weight λ_f
- The couple chooses e_f and e_m to solve

$$\max \left[\log(c) + \tilde{\delta}\left\{\alpha\theta\log(e_f) + (1-\alpha)\theta\log(e_m)\right\}\right]$$

where $\tilde{\delta} \equiv \lambda_f\delta_f + (1-\lambda_f)\delta_m$.
subject to

$$\begin{aligned} 1 &= l_f + e_m + e_f \\ c &= A(l_f H_f)^{\alpha} H_m^{1-\alpha} \end{aligned}$$

- The growth rate of human capital is

$$\frac{H'}{H} = \left\{\frac{B\tilde{\delta}\theta}{\alpha + \tilde{\delta}\theta}(1-\alpha)^{1-\beta}\alpha^{\beta}\right\}^{\theta} \equiv 1 + g^{exog}$$

where $\tilde{\delta} \equiv \lambda_f\delta_f + (1-\lambda_f)\delta_m$

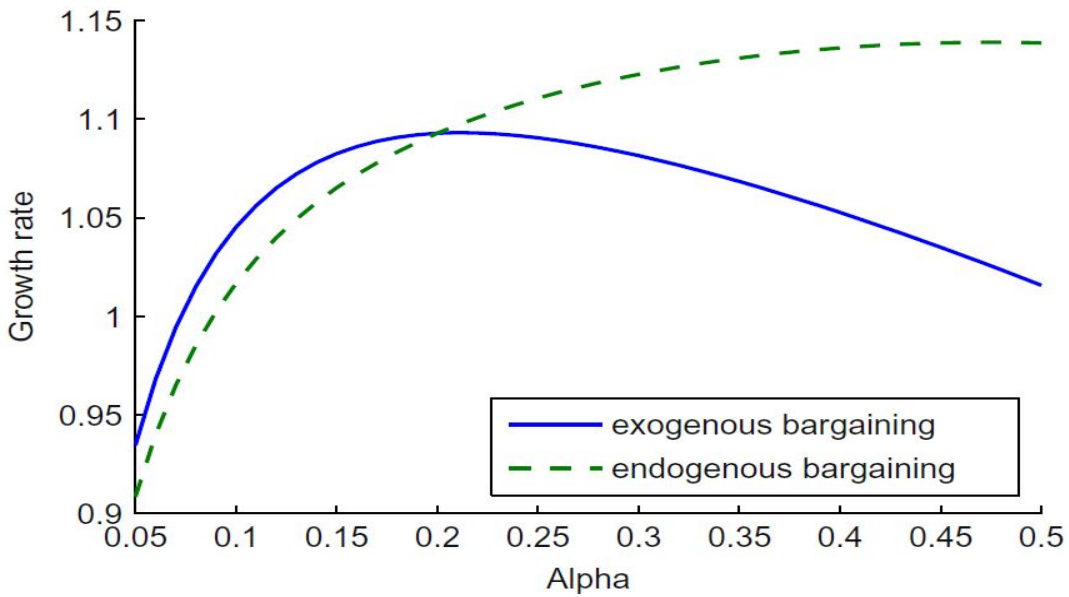
A Model with Two Parents (3) Endogenous Bargaining

- Now assume that the bargaining weight is a function of relative education: $\lambda(e_f, e_m) = \frac{e_f}{e_f + e_m}$
- Recall that relative education is a function of relative importance of female labor: $\frac{e_f}{e_f + e_m} = \alpha$
- The growth rate of human capital is

$$\frac{H'}{H} = \left\{ \frac{B\hat{\delta}\theta}{\alpha + \hat{\delta}\theta} (1-\alpha)^{1-\beta} \alpha^{\beta} \right\}^{\theta} \equiv 1 + g^{end}$$

  where $\hat{\delta} \equiv [\alpha\delta_f + (1-\alpha)\delta_m]$
- Remarks
  - Assume $\delta_f > \delta_m$ (women care about children more than men). Can show that if $\lambda_f < \alpha$, then $g^{end} > g^{exog}$. If $\lambda_f > \alpha$, then $g^{end} < g^{exog}$.
  - When women's power is low relative to their importance in production, endogenizing the link from education to bargaining power increases the growth rate.

A Model with Two Parents (3) Endogenous Bargaining



Numerical example (Figure 23)

- In the model of exogenous λ_f, when α is too high, education cost of children was too high (relative to market work) and growth falls.
- With endogenous λ, however, this effect is mitigated since a rise in α raises the bargaining power of women (assumed to be more altruistic than men), affects family resource allocation to more education and increases human capital.